

1 a)

We know that the Energy of the Ising model is given by:

$$\mathcal{E} = -J \sum_{\langle i,j \rangle} S_i S_j$$

for the 1-D case the neighbours can be simply expressed as:

$$\mathcal{E} = -J \sum_{i=1}^{N-1} S_i S_{i+1}$$

The partition function Z is given by

$$Z = \sum_{S_i} e^{-\beta \mathcal{E}_{S_i}}$$

$$\Rightarrow Z = \sum_{S_i} e^{\beta J \sum_{i=1}^{N-1} S_i S_{i+1}}$$

Using the properties of exponential functions we can write the following:

$$e^{\sum_i a_i} = \prod_i e^{a_i}$$

$$Z = \sum_{S_i} \prod_i e^{\beta J (S_i S_{i+1})}$$

We know that $S_i = \pm 1 \Rightarrow$ given the possible combinations we have:

$S_i =$	1	-1
S_2	1	-1
1	-1	1
-1	1	-1

So we can define the transfer matrix $T_{s,s'}$

$$T_{s,s'} = e^{\beta J s s'}$$

Using the possible values for s, s' , we can write $T_{s,s'}$ in matrix form

$$T = \begin{pmatrix} e^{\beta J} & e^{-\beta J} \\ e^{-\beta J} & e^{\beta J} \end{pmatrix}$$

$$\Rightarrow Z = \sum_{s_i} \prod_i \underbrace{e^{\beta J s_i s_{i+1}}}_{T_{s_i, s_{i+1}}} = \sum_{s_i} \prod_i T_{s_i, s_{i+1}}$$

$$\Rightarrow Z = \sum_{s_i} T_{s_1, s_2} T_{s_2, s_3} \cdots T_{s_{N-1}, s_N}$$

$$Z = \sum_{s_1, s_N} (T^{N-1})_{s_1, s_N}$$

We can further simplify this into

$$Z = (1, 1) T^{N-1} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Assuming T is a diagonalizable matrix we can write

$$T = P D P^{-1}$$

where $D = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$

$$\Rightarrow T^{N-1} = P D^{N-1} P^{-1}$$

The eigen vectors of T are $\vec{\lambda}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ & $\vec{\lambda}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$, associated with the eigenvalues $\lambda_1 = 2 \cosh(\beta J)$, $\lambda_2 = 2 \sinh(\beta J)$

Plugging this into the previous equation gives:

$$Z = 2^N \cosh^{N-1}(\beta J)$$

b)

$$U = - \frac{\partial}{\partial \beta} (\ln(Z)) = - \frac{1}{Z} \frac{\partial}{\partial \beta} (Z)$$

$$\Rightarrow U = - \frac{1}{Z} \frac{\partial}{\partial \beta} \left(\cosh^{N-1}(\beta J) \right) \cdot 2^N$$

$$(N-1) \cosh^{N-2}(\beta J) \cdot \frac{\partial}{\partial \beta} \left(\cosh(\beta J) \right) \\ J \sinh(\beta J)$$

$$\Rightarrow U = -\frac{1}{2} 2^n J(n-1) \cosh^{n-2}(\beta J) \sinh(\beta J)$$

$$U = -\frac{2^n \cosh^{n-2}(\beta J)}{2^n \cosh^{n-1}(\beta J)} J(n-1) \sinh(\beta J)$$

$$U = -\cosh^{-1}(\beta J) J(n-1) \sinh(\beta J)$$

$$U = J(1-n) \tanh(\beta J)$$

$$\beta = \frac{1}{k_B T} \Rightarrow U = J(1-n) \tanh\left(\frac{J}{k_B T}\right)$$

The specific heat is.

$$C = \frac{dU}{dT} = \frac{\partial U}{\partial \beta} \frac{d\beta}{dT}$$

$$C = \frac{\partial}{\partial \beta} \left(J(1-n) \tanh(\beta J) \right) \frac{d}{dT} \left(\frac{1}{k_B T} \right)$$

$$J^2(1-n) \operatorname{sech}^2(\beta J) \cdot -\frac{1}{k_B T^2}$$

$$C = \frac{(n-1)}{k_B} \left(\frac{J \operatorname{sech}\left(\frac{J}{k_B T}\right)}{T} \right)^2$$

c)

$$u = \frac{U}{N} = \frac{(1-N)}{N} J \tanh(\beta J)$$

$$\quad \quad \quad \underbrace{\quad}_{\left(\frac{1}{N} - 1\right)}$$

$$u = \left(\frac{1}{N} - 1\right) J \tanh(\beta J)$$

Similarly for $\frac{C}{N} = c$

$$c = \left(1 - \frac{1}{N}\right) \frac{1}{k_B} \left(\frac{J \operatorname{sech}(\beta J)}{T}\right)^2$$

As $N \rightarrow \infty$, they respectively approach:

$$u = -J \tanh(\beta J) \quad \& \quad c = \frac{1}{k_B} \left(\frac{J \operatorname{sech}(\beta J)}{T}\right)^2$$

or

$$c = \frac{1}{k_B} \left(\frac{\beta}{k_B} J \operatorname{sech}(\beta J)\right)^2$$

$$c = \frac{\beta^2 J^2 \operatorname{sech}^2(\beta J)}{k_B^3}$$

2. a)

We can see that u approaches a constant value as temperature decreases i.e. $\frac{1}{k_B T}$ increases

The constant value " u " approaches has a clear correlation to the number of spins " N ".

For " c " we can see that the maximal value is around $\frac{1}{k_B T} = 1$, i.e. $T = \frac{1}{k_B}$; " c " being close to zero otherwise

We can also see the maximal value depends on N .

2. b)

Looking at the graphs we can see that as N grows the difference between the lines decreases, ~~showing~~ indicating that as $N \rightarrow \infty$, the functions will converge into a fixed function.